

LeetCode 752: Open the Lock

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C++ Solution

```
class Solution {
public:
    int openLock(vector<string>& deadends, string target) {

        unordered_set<string> ded;
        unordered_set<string> seen;
        queue<pair<string,int>> q;
        for(string s:deadends){
            ded.insert(s);
        }

        if(ded.find("0000")!=ded.end()) return -1;

        q.push({"0000",0});
        seen.insert("0000");

        while(!q.empty()){
            string currs=q.front().first;
            int currd=q.front().second;

            q.pop();
            if(currs==target) return currd;

            for(int i=0;i<4;i++){
                string news=currs;
                news[i]++;
                if(news[i]>'9'){
                    news[i]='0';
                }

                if(seen.find(news)==seen.end() && ded.find(news)==ded.end()){
                    seen.insert(news);
                    q.push({news,currd+1});
                }

                news[i]--;
                if(news[i]<'0'){
                    news[i]='9';
                }

                if(seen.find(news)==seen.end() && ded.find(news)==ded.end()){
                    seen.insert(news);
                    q.push({news,currd+1});
                }
            }
        }

        //ab ek baar peeche ghuma ke dekhenge
    }
};
```

```

        news=currs;
        news[i]--;
        if(news[i]<'0'){
            news[i]='9';
        }

        if(seen.find(news)==seen.end() && ded.find(news)==ded.end()){
            seen.insert(news);
            q.push({news, currd+1});
        }
    }
}

return -1;

}
};

```

Algorithm & Notes

store all deadends in a set.

If "0000" is a deadend, return -1.

Use BFS because every rotation costs 1.

Queue stores: **{current_state, moves}**

Start BFS from: **{"0000", 0}**

For every popped state:
If it is the target, return moves.

Generate 8 neighbours:
Increase each of 4 digits.
Decrease each of 4 digits.

Handle rotation:

9 -> 0

0 -> 9

If neighbour is:

not in deadends

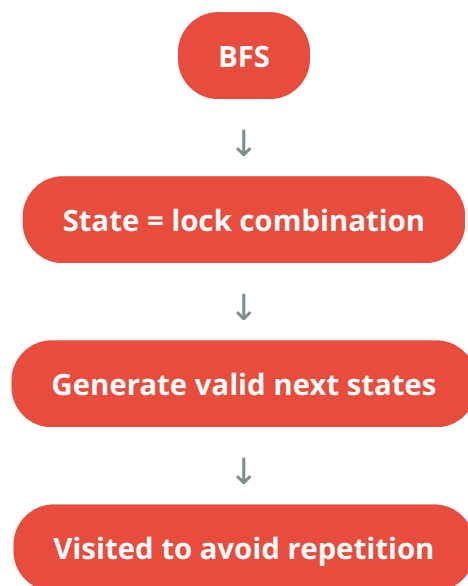
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push with moves + 1.

If BFS finishes without reaching target, return -1.

Pattern & Strategy



asked in zepto , deshaw , JPMC